

Education

University of Illinois Urbana-Champaign Ph.D. Candidate in Electrical & Computer Engineering Advisor: Prof. Klara Nahrstedt	GPA: 3.73 / 4.0	Sep. 2024 – May 2028 (expected)
Tsinghua University M.E. in Artificial Intelligence Advisor: Prof. Haoqian Wang	GPA: 3.63 / 4.0	Sep. 2021 – Jun. 2024
Dalian Maritime University B.E. in Automation Rank: 1 of 123	GPA: 4.22 / 5.0	Sep. 2017 – Jun. 2021

Academic Research

Embodied Multi-Agent Planning • Developing and testing planners for embodied multi-agent collaboration on the PARTNR benchmark.	Ongoing
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Streaming Video Understanding • Developed StreamClarify, a training-free streaming video QA framework for frozen vision-language models (VLMs), using entropy-based clarification, temporal-pyramid key-value caching, and semantic retrieval. • Built StreamClarify-QA with 1,837 QA pairs from 355 videos and evaluated three VLM backbones. Improved macro accuracy by up to 8.2 percentage points over existing streaming video QA methods.
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Computer Vision for Medical Imaging • Developed Transformer-based multiple instance learning methods for weakly supervised whole-slide image classification and survival prediction, using spatial context and noisy pseudo-labels. • Applied diffusion models to feature augmentation and semi-supervised cell segmentation, using unsupervised pre-training, collaborative learning, and content-guided sampling. • Studied task-informed image reconstruction with a data-consistent U-Net and null-space learning under task and observer shifts.

Industry Research

Applied Scientist Intern · Amazon Alexa AI (Multimodal Agents) • Developed Tab Zero, a unified browser interface that lets web agents inspect egocentric video and complete web tasks using shared browser actions. • Built and tested a distributed reinforcement learning pipeline with GRPO, SGLang, Megatron-LM, and parallel Chromium rollouts. Tested rule-based and VLM-based rewards. • Evaluated three model families on Ego2Web and LongEval (3–20-minute videos). Improved zero-shot task success rate by up to 8.0 percentage points over uniform sampling on the filtered Ego2Web subset.	May 2026 – Aug. 2026
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Honors & Skills

National Scholarship (top 1%), Tsinghua University	2023
National Scholarship (top 1%), Dalian Maritime University	2018, 2019, 2020
Programming & Tools: Python, PyTorch, LaTeX	
Languages: English (fluent), Mandarin (native)	

Publications

Streaming Video Understanding & Multimodal Agents

1. **StreamClarify: Proactive Clarification for Grounded Streaming Video Question Answering.** Zhuchen Shao, Bo Chen, Klara Nahrstedt. *Under review*
2. **Tab Zero: Grounding Web Agents in Egocentric Video.** Zhuchen Shao, et al. *In preparation*

Computer Vision for Medical Imaging

3. **Investigating the impact of data consistency in task-informed learned image reconstruction method.** Zhuchen Shao, Changjie Lu, Kaiyan Li, Hua Li, Mark A. Anastasio. *SPIE Medical Imaging, 2025 [Paper]*
4. **Prior-guided diffusion model for cell-level segmentation in quantitative phase imaging.** Zhuchen Shao, Mark A. Anastasio, Hua Li. *SPIE Medical Imaging, 2025 [Paper]*
5. **Semi-supervised semantic segmentation of cell nuclei with diffusion model and collaborative learning.** Zhuchen Shao, Sourya Sengupta, Mark A. Anastasio, Hua Li. *Journal of Medical Imaging, 2025 [Paper]*
6. **AugDiff: Diffusion based Feature Augmentation for Multiple Instance Learning in Whole Slide Image.** Zhuchen Shao, Liuxi Dai, Yifeng Wang, Haoqian Wang, Yongbing Zhang. *IEEE Transactions on Artificial Intelligence, 2024 [Paper]*
7. **LNPL-MIL: Learning from Noisy Pseudo Labels for Promoting Multiple Instance Learning in Whole Slide Image.** Zhuchen Shao, Yifeng Wang, Yang Chen, Hao Bian, Shaohui Liu, Haoqian Wang, Yongbing Zhang. *ICCV, 2023 [Paper]*
8. **HVTSurv: Hierarchical Vision Transformer for Patient-level Survival Prediction from Whole Slide Image.** Zhuchen Shao, Yang Chen, Hao Bian, Jian Zhang, Guojun Liu, Yongbing Zhang. *AAAI, 2023 [Paper]*
9. **dMIL-Transformer: Multiple Instance Learning via Integrating Morphological and Spatial Information for Lymph Node Metastasis Classification.** Yang Chen*, Zhuchen Shao*, Hao Bian*, Zijie Fang, Yifeng Wang, Yuanhao Cai, Haoqian Wang, Guojun Liu, Xi Li, Yongbing Zhang. *IEEE Journal of Biomedical and Health Informatics, 2023 [Paper]*
10. **Multiple Instance Learning with Mixed Supervision in Gleason Grading.** Hao Bian*, Zhuchen Shao*, Yang Chen*, Yifeng Wang, Haoqian Wang, Jian Zhang, Yongbing Zhang. *MICCAI, 2022 [Paper]*
11. **TransMIL: Transformer based Correlated Multiple Instance Learning for Whole Slide Image Classification.** Zhuchen Shao*, Hao Bian*, Yang Chen*, Yifeng Wang, Jian Zhang, Xiangyang Ji, Yongbing Zhang. *NeurIPS, 2021 [Paper]*

* Equal contribution.